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Automated fire risk assessment and mitigation in building blueprints using computer vision and deep generative models

Dayou Chen [a], Long Chen [a,*], Yu Zhang [a], Shan Lin [b], Mao Ye [c], Simon Sølvsten [d]

[a] School of Architecture, Building and Civil Engineering, Loughborough University, United Kingdom
[b] Guangzhou Metro Design & Research Institute Co. Ltd., 510010, Guangzhou, China
[c] Wuhan Zhongyuan Electronic Group Co., Ltd., China
[d] European Center for Risk & Resilience Studies, University of Southern Denmark, Denmark

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ABSTRACT

Building fire risks pose significant threats to individual safety and bear substantial economic consequences. Consequently, developing effective automated fire risk assessment and mitigation solutions has become increasingly crucial to mitigate fire risks and reduce losses. Previous automated fire risk assessment approaches have predominantly relied on structured building design information, such as Industry Foundation Classes (IFC)-based Building Information Modelling (BIM) models, which limited their applicability in scenarios without such data. Additionally, there is a notable absence of a comprehensive approach in existing research for effectively mitigating fire risks identified during the assessment process. This study aims to bridge these gaps by proposing an innovative approach for assessing and mitigating fire risks using raw building blueprints. This approach incorporates advanced computer vision techniques to process both paper-based and digital blueprints. It then employs a knowledge-based algorithm for evaluating fire safety and regulatory compliance within these blueprints. A key innovation is the development of a deep generative model that redesigns unqualified blueprint designs to meet safety standards. This research contributes to the field by providing a more capable, accessible, and flexible approach for automated building safety, introducing Artificial Intelligence (AI)-enabled solutions for risk mitigation. It offers a versatile option applicable to various building types, significantly enhancing fire safety and compliance, especially for buildings without extensive BIM data. This study addresses the limitations of current methodologies and lays the groundwork for future advancements in automated fire risk assessment and mitigation.

1. Introduction

The devastating impact of building fires remains a consistent issue in societies, threatening both lives and property across urban and rural areas. The National Fire Protection Association reported that in 2021, fires in the United States resulted in a staggering $15.9 billion in property damage [1], highlighting the immense economic toll. Besides financial losses, the tragic Grenfell Tower fire in 2017 exemplifies the dire threat to human lives [2], underscoring the profound need for effective fire risk identification and reduction measures.

Considerable endeavours have been undertaken to evaluate and address fire risks in buildings. According to ISO 31000:2009, fire risk assessment includes three main stages: risk identification, risk analysis and risk evaluation [3]. The Dow Corning Company proposed the first modern fire risk assessment method in the US in the 1970s, in which they applied the fire and explosion index assessment method to conduct a systematic fire risk assessment of fire risk for enterprises [4]. Afterwards, various similar solutions subsequently released, including the Gretener Method by M. Gretener from Switzerland [5], and FiRECAM by the National Research Council of Canada (NRC) [6], have set the foundation for evaluating fire risks. However, these manual processes are prone to errors, time-consuming, and costly [7], creating a demand for more efficient, automated solutions.

In recent efforts to automate fire risk assessment, Malsane et al. [8] developed a semantically rich object model for automating building code compliance checks, particularly focusing on fire safety in residential houses using Industry Foundation Classes (IFC)-based Building Information Modelling (BIM) data. Jiang et al. [9] proposed a multi-ontology fusion methodology integrating ontologies with rule-based reasoning for automated code compliance. This approach combines a code ontology with a model ontology using a grey-box technique

* Corresponding author.
E-mail address: l.chen3@lboro.ac.uk (L. Chen).

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to streamline the checking process. Additionally, Fitkau and Hartmann [10] introduced a detailed ontology model for automatic compliance checking of structural fire safety requirements using existing BIM models. These approaches, however, often require structured building design information, such as IFC-based BIM models, limiting their application in many scenarios without such data. Another notable gap in current automated fire risk assessment research is the absence of a comprehensive methodical approach for mitigating the fire risks identified.

In the field of automated fire risk mitigation, significant focus has been on post-event hazard detection and evacuation strategy development. Cheng et al. [11] developed a graph-based network to evaluate the shortest evacuation routes in a building using live CCTV (Closed-Circuit Television) camera footage. Meanwhile, Li et al. [12] developed a dynamic intelligent evacuation model that considers fire propagation to determine the most effective evacuation strategy in the event of a fire in a building. However, few proactive studies that incorporate mitigation strategies into automated fire risk assessments to improve fire safety.

Fire safety design is increasingly recognised as a critical method for fire risk prevention and mitigation. Maluk et al. [13] explored the potential benefits of incorporating fire safety into the building design process, as it is often viewed as an additional constraint rather than a design parameter in current design practice. Mazzucchelli et al. [14] proposed performance-based fire safety design that allows for customised design based on specific project requirements. Sabbaghzadeh et al. [15] developed a framework to obtain appropriate fire safety measures while considering their effects on safe evacuation and budget limits, which can help decrease fatalities during fires.

A key factor for both fire risk assessment and fire safety design is the consideration of evacuation distance. This consideration is key to minimising response time and enhancing safety, as rapid evacuation is critically important in hazardous situations. Intuitively, shorter routes ensure swifter and safer evacuations [16]. Such distances are regulated worldwide, for consistency, this research refers such a distance as the Maximum Travel Distance (MTD). For instance, in public buildings in China, a minimum distance of 25 metres is stipulated for care facilities [17]. The German Federal Institute for Occupational Safety and Health (BAuA) [18] mandates that primary escape routes should not exceed 35 metres. The UK offers more nuanced specifications based on the type and conditions of dwellings and flats [19]. Accurately determining MTD from building blueprints is essential for assessing fire risks and is a key focus of this research.

This research aims to pioneer an innovative approach based on Artificial Intelligence (AI) techniques for assessing and proactively mitigating fire risks utilising raw building blueprints. This approach will not only assess fire risks but also integrate mitigation strategies from the initial steps, using MTD as a crucial parameter for fire safety design. This methodology encompasses three pivotal steps:

- Automated polymorphic building blueprints standardisation and interpretation, which will standardise both paper-based and digital building blueprints and automatically interpret the blueprints using computer vision.
- Automated building fire risk assessment, which will assess building fire risks and ensure compliance with established fire regulations.
- AI-enabled building fire risk mitigation, which will integrate AI and deep generative models to mitigate identified fire risks.

The research scope is centred on applying these methodologies to building layout design blueprints. This design data representation is less structured but more widely available compared to detailed BIM models. This focus addresses a critical gap in automatic fire risk evaluation for many buildings without BIM data, enhancing the accessibility and flexibility of fire risk assessments and mitigations.

The procedure is initiated by pre-processing to standardise both paper-based and digital building blueprints. This initial step is crucial as it allows the application of advanced AI techniques to a broader range of building types, many of which may only have blueprint design documentation available. This is followed by a comprehensive interpretation of building blueprints. This interpretation involves segmenting rooms, obtaining room-specific information, extracting fire resistance levels, and vectorising room layouts. Subsequently, the interconnectivity of each spatial segment is analysed for fire risk assessment, such as determining the MTD, followed by ascertaining compliance with fire safety regulations. It is important to note that while the automated assessment is rigorous, it is specifically designed to focus on the geometric and topological aspects of design, particularly the MTD criterion, as stipulated in the UK Building Regulations. Suppose specific building designs are found non-compliant with fire safety norms. In that case, their layout undergoes redesign and modification using AI, specifically, a connectivity graph guided deep generative model, until they satisfy fire risk evaluation standards. This approach provides a proactively effective tool for designers and engineers to enhance the fire safety of building layouts based on building blueprints, at the design and retrofitting phases.

This rest of this paper is structured as follows: Section 2 presents a comprehensive literature review, laying the foundation for understanding the current state of AI in building blueprint interpretation, design and risk compliance. Section 3 elucidates the conceptual basis and the detailed methodologies of each pivotal step of the proposed approach. Section 4 elaborates a close implementation of the proposed approach, forming a case study. Section 5 thoroughly discuss the approach's capabilities and its comparison with existing studies. The paper concludes with a summary of findings and prospects for future research in Section 6.

2. Literature review

This review begins by exploring the current methodologies and technologies utilised in the automated interpretation of building blueprints, emphasising the application of computer vision techniques for analysing blueprints. This is followed by examining existing approaches to automated building fire risk assessment and compliance checking, mainly focusing on previous methods of automated compliance related to fire safety. Finally, attention is turned towards using AI in building layout design, highlighting the potential of deep generative AI models for fire risk mitigation strategies.

2.1. Automated interpretation of building blueprints

In the domains of Architecture, Engineering, and Construction (AEC), as well as in the realm of facility management and property management, building blueprints serve as the bedrock of information, dictating design, construction, and maintenance processes. The transition from traditional paper-based blueprints to digital formats, primarily driven by advancements in Computer-Aided Design (CAD), represents a pivotal shift in how this information is accessed, analysed, and utilised. However, this transition is not uniform across industries, regions and countries. A significant portion of existing buildings, particularly those constructed before the digital era, continue to rely on blueprints in less accessible formats, such as physical paper copies or non-interactive PDF files. This disparity presents a unique challenge to uniformly interpret and utilise building blueprints of varying formats for critical tasks in fire risk assessment and compliance checking. The interpretation of building blueprints plays a crucial role in this process, as it allows for the identification of potential hazards, the layout of evacuation routes, the location of fire safety facilities, and the overall compliance with building codes and safety standards.

Given the diversity of blueprint formats, from physical paper copies to CAD files, the task of interpreting them for risk assessment becomes complex. Traditional methods, which often rely on manual interpretation by experts, are time-consuming, subject to human error, and not scalable to the vast number of buildings that must be assessed regularly. This is where the field of automated blueprint interpretation, employing techniques from machine learning, AI, computer vision, and Optical Character Recognition (OCR), presents a promising solution. For thorough fire risk analysis and evaluation based on building blueprints, careful interpretation and analysis of these blueprints in various formats are crucial. This involves identifying potential fire hazards and devising appropriate safety measures by processing the extracted information to achieve key objectives: identifying the building layout, locating key passive and active protections, and understanding building design and construction [20]. By automating the interpretation of blueprints, these technologies aim to bring efficiency, accuracy, and consistency to the process, making fire risk assessment more reliable and accessible.

2.1.1. CAD Conversion/Digitalisation

The digitalisation of building blueprints, transitioning from traditional physical formats to versatile, machine-readable CAD formats, was not just a technical evolution but a paradigm shift in architectural data management. This transition is pivotal for enabling automated processing, interpretation, and analysis of building blueprints, which are foundational for effective fire risk assessment and compliance checking. The challenge here is multifaceted, involving not only the conversion of diverse blueprint formats but also the preservation and interpretation of intricate details and metadata crucial for fire safety analysis.

Early efforts in CAD conversion and digitalisation focused on digitising paper-based and hand-drawn architectural plans. Lladós et al. [21] developed a system for processing hand-drawn architectural drawings within a CAD framework, focusing on identifying building elements and their spatial relationships using attributed graphs. In a similar vein, Dori and Liu [22] worked on converting mechanical engineering drawings from scans to 3D reconstructions, concentrating on the structural and algorithmic aspects. These works paved the way for a comprehensive understanding and reconstruction of complex engineering and architectural drawings.

Recent advancements have further refined these concepts. Henderson and Swaminatha [23] introduced a structural modelling approach combined with a non-deterministic agent system for interpreting CAD drawings, emphasising the importance of understanding the semantics behind engineering drawings, beyond just capturing geometric data. Tang et al. [24] developed a system specifically for analysing and labelling structural scenes in CAD format, employing advanced strategies for detecting and recognising various components in floorplan drawings. These developments demonstrate significant progress in the capability to digitalise and interpret complex architectural plans effectively.

The advancements in CAD conversion and digitalisation have profound implications for fire risk assessment and compliance checking. The ability to convert and interpret diverse blueprint formats into a standardised digital format is crucial for accurately identifying potential fire hazards, planning effective evacuation routes, and ensuring comprehensive compliance with fire safety standards.

2.1.2. Room classification

Classifying rooms within a building blueprint is paramount for interpreting blueprints and automated fire risk assessment. The classification determines not only the usage and function of each space but also informs the specific fire safety standards and requirements that apply. This aspect of blueprint interpretation is especially challenging due to the diverse nature of room types and the subtleties involved in distinguishing them based on layout and design features.

Dehbi et al. [25] focused on developing a classifier to predict the functional use of housing rooms. Their approach utilised features like room areas and orientation, extracted from an extensive database of annotated rooms. This approach shifted room classification from manual to automated, data-driven methods. Following this, Mewada et al. [26] proposed an innovative floorplan information retrieval algorithm based on shape extraction and room identification. This work further advanced room classification by leveraging geometric analysis, allowing for more nuanced differentiation between room types, critical for tailoring fire safety measures to specific room functions and designs.

More recently, Paudel et al. [27] explored using graph neural networks for room classification on building floorplan. By representing floorplans as undirected graphs, their method demonstrated substantial accuracy improvements over baseline deep learning methods. This approach showcased the effectiveness of advanced AI techniques in capturing the intricate relationships and patterns within architectural layouts, crucial for accurate room classification.

These developments in room classification technology are crucial for enhancing fire risk assessment, allowing for precise application of fire safety standards and effective evacuation strategy planning based on room functions. The move towards AI-driven methods marks significant progress in understanding the intricacies of architectural design for safety and compliance purposes.

2.1.3. Information and knowledge extraction

The extraction of detailed and accurate information from building blueprints is fundamental in automated blueprint interpretation for fire risk assessment and compliance checking. This involves not just recognising architectural elements and room classifications but also understanding deeper layers of information such as material specifications, fire safety facilities, and other semantic details that are crucial for a thorough fire safety analysis.

Initial developments in this area were primarily rule-based systems focused on keywords. For instance, Hirasawa et al.'s [28] work demonstrated the potential of CAD systems in intelligent information processing, particularly in systematising knowledge like technical standard reviews and constructibility investigations in construction projects. Following this, Lu et al. [29] introduced a method for automated knowledge extraction from engineering table drawings, proving significant for extracting safety-critical information from complex engineering drawings.

A notable advancement in this field came with the application of deep learning techniques. Schönfelder et al. [30] proposed a deep learning-based method for detecting and classifying text elements in legacy architectural drawings. This approach is particularly relevant for automating the extraction of room stamps and textual data from floorplans, thereby facilitating the enrichment of digital twins of existing structures with detailed and accurate information.

The progression in information and knowledge extraction technologies is pivotal for enhancing the accuracy and efficiency of fire risk assessments. By automating the process of extracting detailed and relevant information from blueprints, these technologies reduce the risk of human error and ensure a more comprehensive understanding of the building's characteristics and potential fire hazards.

2.1.4. Floorplan vectorisation

Floorplan vectorisation represents a vital component in the automated interpretation of building blueprints, especially for fire risk assessment and compliance checking. This process involves converting rasterised images of blueprints into vector graphics, which are more conducive to machine processing and analysis. Vectorisation not only simplifies the representation of complex building layouts but also ensures that the geometric and topological information of buildings is accurately captured in a format that can be readily analysed by automated systems.

The field of floorplan vectorisation began with significant contributions like Liu et al.'s [31] learning-based approach to convert

Table 1
Summary of key areas in automated interpretation of building blueprints.

Area	Key contributions	Implication
CAD Conversion/Digitalisation	Early systems for understanding hand-drawn architectural drawings [21]; Development of 3D CAD models from mechanical drawings [22]; Structural modelling for CAD interpretation [23]; Automatic labelling of CAD floorplans [24].	Enhanced ability to process diverse blueprint formats
Room Classification	Classifier for functional use of housing rooms [25]; Floorplan information retrieval algorithm [26]; Graph neural networks for room classification [27].	Identification of room types for fire safety standards
Information and Knowledge Extraction	CAD systems for knowledge processing in construction [28]; Automated analysis of engineering drawing layouts [29]; Deep learning for text classification in architectural drawings [30].	Enables extraction of crucial safety information and details from blueprints
Floorplan Vectorisation	Conversion of rasterised images to vector graphics [31]; Embedding floorplans as vectors using attributed graphs [32]; 2D to 3D vector data conversion [33]; Vectorising line drawings for topology [34].	Provides detailed and accurate representations of building layouts

rasterised floorplan images into vector graphics. Their method employed a neural architecture to transform an image into a set of junctions, representing low-level geometric and semantic information. Their method was pivotal in creating vectorised floorplans that could be easily processed by automated systems. Following this, Azizi et al. [32] presented a technique for embedding floorplans as vectors using an attributed graph. This approach incorporated geometric information, design semantics, and behavioural features, utilising Long short-term memory (LSTM) and Variational Autoencoder (VAE) architectures to generate meaningful and accurate vector representations for floorplans. This technique allowed for a richer representation that included both physical layout and functional aspects of spaces.

Recent advancements have expanded the scope of vectorisation. Park and Kim [33] developed a method for converting 2D raster drawings into 3D vector models, enhancing the understanding of building structures. Similarly, Zhang et al. [34] introduced a method for vectorising line drawings of varying thicknesses, focusing on reconstructing their topology, crucial for interpreting complex architectural details accurately.

These developments in floorplan vectorisation have significantly improved the ability to conduct thorough fire risk assessments and compliance checks. By providing accurate and detailed building layout representations, these technologies facilitate a more comprehensive analysis of potential fire hazards and safety measures.

2.1.5. Summary

As summarised in Table 1, this review has traversed the landscape of automated interpretation of building blueprints. Four critical areas: CAD Conversion/Digitalisation, Room Classification, Information and Knowledge Extraction, and Floorplan Vectorisation were explored. Each of these areas represents a fundamental component in the complex process of blueprint interpretation, and their collective advancements depict a rapidly evolving field driven by technological innovation.

A key trend observed across all categories is the increasing reliance on advanced AI and machine learning techniques. These technologies are not only enhancing the accuracy and efficiency of interpreting blueprints but are also enabling the processing of more complex and data-rich building representations. However, these technological advancements also highlight a gap, i.e., the need for standardised protocols and comprehensive approach to integrate these diverse technologies into a cohesive system for fire risk assessment and compliance checking.

2.2. Automated building fire risk assessment and compliance checking

2.2.1. Building regulations on fire safety evacuation requirements

As rapid and safe evacuation is pivotal in minimising fire hazards, one of the cornerstones of fire safety design is the management of evacuation routes, specifically the distances people must travel to reach safety. Examining how different countries regulate means of escape and evacuation travel distances provides valuable insights into how fire risks may be assessed and mitigated.

Table 2 summarises the fire safety regulations regarding fire safety evacuation from four major countries: Australia, China, Italy, and the UK. These regulations were selected due to their public accessibility and representativeness of broader practices across their respective continents and regions. These regulations outline the design and safety features necessary for residential buildings, focusing on the criteria for escape routes and travel distance requirements. The National Construction Code (NCC) 2019 Volume One Amendment 1 [35] of Australia requires two exits for buildings taller than 25 metres or those exceeding six stories. Specifically, it details exit travel distances for Class 2 and Class 3 buildings, which encompass the most common residential building types. The regulations stipulate that the entrance doorway of any sole-occupancy unit must be no more than 6 metres from an exit or from a point from which travel in different directions to 2 exits is available. Alternatively, it can be up to 20 metres from a single exit that leads directly to a road or open space. Moreover, any area within a room that is not a sole-occupancy unit must not exceed 20 metres from an exit or from a point at which travel in different directions to 2 exits is available. According to the Code for Fire Protection Design of Buildings (GB 50016 - 2014) [17] of China, two exits are required for buildings with an effective height of over 54 m. The MTD varies with fire resistance levels, ranging from 15 to 22 metres for any point to flat door and 25 to 40 metres for flat door to stair lobby, increasing with the fire resistance levels of the building. The Technical Fire Prevention Standards [36] of Italy require two exits in most buildings; in buildings where occupants exceed 150 with a high-risk profile or over 500 in general, three exits are stipulated. The document specifies that the travel distances should not exceed 20 to 70 metres depending on the risk category determined by the reference risk profile. The UK Building Regulations 2010, Approved Document B (Fire Safety) Volume 1: Dwellings (2019) [19], specifies the requirements for means of escape in residential buildings. It allows a single escape route from the flat entrance door under certain conditions, such as the flat being on a storey served by a single common stair with protective measures. The MTD from any point to the flat entrance door in a habitable room must not exceed 9 m, and from the flat door to the stair lobby, the distances can vary from 7.5 to 30 m, depending on the number of escape routes available. The UK regulation has been adopted as a basis for the criteria for the proposed automated fire risk assessment approach due to its comprehensive guidelines for residential buildings and because a significant portion of the blueprint data used in this study is sourced from the UK. However, the proposed approach is flexible and can be readily extended to accommodate compliance checking under different national codes.

Table 2
International building regulations on fire safety evacuation requirements.

Country	Number of means of escape	Escape travel distances
Australia	2 exits required for: (a) effective height > 25 m (b) more than 6 stories	From unit to entrance doorway < 20 m
China	2 exits required for: (a) effective height > 54 m (b) 54 m > effective height > 27 m and area > 650 m^2 and any distance from flat door to stair > 10 m (c) effective height < 27 m and area > 650 m^2 and any distance from flat door to stair > 15 m	From any point to flat door < 15 m/20 m/22 m (varying by fire resistance level) From flat door to stair lobby < 25 m/35 m/40 m (varying by fire resistance level)
Italy	3 exits required for: (a) Occupants > 150 and high risk profile (b) Occupants > 500 most other cases: 2 exits	Escape routes and travel distances ranging from 20 m to 70 m based on risk profile
UK	Single escape route is acceptable under specific conditions such as served with single common stair with protective measures; otherwise, two exits required	From any point to flat door < 9 m From flat door to stair lobby < 7.5 m/30 m (escape in one direction/ in more than one direction)

Overall, the number of required evacuation exits typically ranges from one to three depending on the storey heights and occupancy levels, whereas the MTDs are distinctly set by each country to enhance safety and minimise response time during evacuations. Although the specific distances and requirements vary, the universal aim of these regulations is to keep travel distances as short as possible to facilitate rapid and safe evacuations in the event of a fire.

2.2.2. Automated code compliance checking

Building fire risk assessment and compliance checking is a complex process that often requires years of expertise, an in-depth understanding of the national building codes and manual labour. These critical processes evaluate adherence to fire safety standards, pinpoint potential area of risks requiring attention, and are pivotal in safeguarding buildings against fire hazards [37–40]. Research has underscored the negligence of fire safety preparedness in healthcare facilities, revealing shortcomings in equipment, systems, and staff awareness [37]. Similarly, educational institutions frequently exhibit non-compliance with fire codes, manifested through inadequate knowledge, absence of dedicated fire response teams, and disregard for regulatory mandates [41]. Comprehensive audits are imperative for ensuring conformity with fire protection regulations, scrutinising aspects like evacuation protocols, smoke barriers, and the fire resistance of building components [42]. Although these measures are instrumental in mitigating fire risks, safeguarding occupants, and enhancing a building's overall fire safety profile, the manual approaches to building fire risk assessment and compliance checking are notably error-prone, time-consuming and costly [7]. Consequently, the automation of building code compliance checks has gained considerable attention as a promising alternative. This process typically include four phases: rule interpretation, building design model preparation for assessment, rule execution and reporting [43].

Early developments of national regulation checking systems like SEUMTER [44], CORENET [45], SMARTcodes [46], and GSA [47] represent foundational efforts in developing systems for automating general building code compliance checking. The checking process is concerned with comparing a building design to the relevant regulations, and Building Information Models (BIMs) provide detailed design information in a computer-readable format. Consequently, most existing automated code compliance checking systems predominantly rely on information extracted from BIM models, typically aligning with the IFC schema [48].

Recent studies in general building automated code compliance have leveraged modern technologies to enhance the efficiency and accuracy of building safety regulations and design evaluations. A significant shift from traditional hard-coded rule-based systems to ontological and semantic rule-based systems has been observed. Malsane et al. [8] developed a semantically rich object model for automating compliance checking, specifically focusing on fire safety in dwelling houses using IFC-based BIM data. Jiang et al. [9] proposed a multi-ontology fusion methodology, which integrates ontologies with rule-based reasoning for automated code compliance. This method combines a code ontology with a designed model ontology using a grey-box technique, streamlining the checking process. Furthermore, Fitkau and Hartmann [10] introduced the Fire Safety Ontology (FiSa) for automatic compliance checking of structural fire safety requirements, emphasising the need for formalising domain knowledge.

The integration of various rules and data sources in automated compliance checking has also emerged as a notable research area, with Ghannad et al. [49] presenting a framework for automated BIM data validation. This framework integrates an open-standard standard, LegalRuleML, pointing to the potential for modularised design evaluation processes in promoting transparency, flexibility, and interoperability. Beach et al. [50] proposed an ecosystem approach to compliance checking, aiming to integrate a variety of software tools and data models, extending beyond BIM data.

Extensive research has also focused on automating the process of capturing building regulations into a computer-readable format. Zhang et al. [51] integrated semantic natural language processing (NLP) with logic reasoning for fully automated code checking. They concentrated on extracting and formalising requirements from codes using NLP and logic-based reasoning. Zhou and El-Gohary [52] used deep learning for semantic information alignment in BIMs, targeting the automation of the alignment of rules and building information for energy compliance checking. Additionally, Zhang and El-Gohary [53] explored the use of natural language generation and deep learning for intelligent building codes, aiming to reduce errors in requirement representation and facilitate intelligent code analytics.

Machine learning and deep learning techniques have also been explored recently in automated compliance checking. Bloch et al. [54] introduced graph-based learning using graph neural networks for automated code checking, representing a shift towards scalable solutions in automated code compliance using deep learning models with attempts to eliminate the need for hard-coded rules.

Table 3 summarises the advancements in utilising modern techniques to improve the automated code compliance checking process. However, the existing literature primarily focused on general code

Table 3
Summary of literature on automated code compliance checking.

Literature	Input requirements	Scope	Technical approach
Ghannad et al. [49]	Building regulation with open-standard schema	General regulation	Visual programming and rule-based reasoning
Bosché [55]	3D scans and structured CAD information	Dimensional compliance	Mathematical model
Nahangi and Haas [56]	3D scans and structured CAD information	Pipe spool fabrication	Rule-based reasoning
Patlakas et al. [57]	Component dimensions and coordinates	Dimensional compliance	Mathematical model
Malsane et al. [8]	IFC-based BIM model	Fire Safety	Rule-based reasoning
Zhang et al. [51]	IFC-based BIM model	General regulation	Natural language processing
Beach et al. [50]	BIM data integrated with software tools	General regulation	Integration with existing platforms
Jiang et al. [9]	IFC-based BIM model	General regulation	Ontology model/ semantic rule-based system
Zhou and El-Gohary [52]	Multiple data representations of BIM models	General regulation	Natural language processing
Xue and Zhang [58]	Semantically annotated building code	General regulation	Natural language processing
Zhang and El-Gohary [53]	Semantically annotated building code	General regulation	Natural language processing
Fitkau and Hartmann [10]	Semi-structured interviews with fire safety planners	Fire safety	Ontology model/ semantic rule-based system
Bloch et al. [54]	Building geometric data provided in any representation	General regulation	Graph neural networks

compliance checking, with a lack of specific emphasis on fire risks. Most systems relied on detailed structured data such as BIM models, limiting their applicability in scenarios where such data is incomplete or unavailable. It highlights the imperative for more exhaustive investigations in automated building fire risk assessment and the necessity to explore alternative methodologies, such as the direct interpretation of building layout blueprints, to enhance understanding and mitigation of building fire risks, particularly through retrofitting strategies for buildings without BIM data. This gap underscores the need for developing systems capable of effective automated code compliance with less structured data, such as 2D CAD drawings, and focusing specifically on fire risk compliance.

2.3. AI for building layout designing

Automated design processes of building layouts have garnered research interest even prior to the advent of deep learning, aimed at enhancing design efficiency and reducing manual labour costs [59,60]. A number of studies have illuminated the potential of deep learning techniques in generating building blueprints. Zhou et al. [61] presented an approach for automatically generative design and optimisation of hospital building layouts. Huang et al. [62] deployed Generative Adversarial Networks (GANs), a subset of deep generative models, for generating floorplans, segmenting various functional areas to facilitate data-driven design. Sharma et al. [63] utilised Convolutional Neural Networks (CNN) and GANs for the detection and retrieval of blueprints. Moreover, Nauata et al. [64] introduced a graph-constrained floorplan generative model grounded in GANs. Jiang et al. [65] proposed a site-embedded generative adversarial network (ESGAN) for automated building layout generation, which encodes a conditional vector to improve performance in different design scenarios and display generated layouts at varying levels of detail based on human–system interaction. Noteworthy is the recent advent of the diffusion model [66] as an emergent technique in deep generative models. Dhariwal and Nichol [67] refined the architecture of diffusion models through classifier guidance, surpassing the performance of GANs in image synthesis. Shabani et al. [68] introduced significant advancements in floorplan design that are achieved with diffusion model, which outperforms state-of-the-art methods on the RPLAN [69] building interior floorplan dataset.

Deep learning methods, particularly deep generative models, have the potential to efficiently design a diverse variety of building blueprints, thereby facilitating enhanced room connectivity and the optimisation of evacuation routes through the reconfiguration of building layouts. However, these methods have not yet been integrated into a risk management workflow, specifically in using AI to redesign blueprints to reduce risks. This integration could represent a significant advancement in the field of fire safety and risk management.

3. Research methodology

This research develops an innovative approach based on AI techniques, depicted in Fig. 1, to aid in building fire risk assessment and proactive mitigation, consisting of three interconnected steps. (1) Step 1, automated polymorphic building blueprint standardisation and interpretation, involves pre-processing polymorphic building blueprints for analysis. This includes resizing, denoising, binarisation, and morphological processing. Computer vision and automatic information extraction techniques are used to identify the structural and functional aspects of buildings. This step generates two outputs: the vectorised floorplan layout and the blueprint metadata that contains material specifications, room type identification, fire safety facilities, and structural dimensions of the building design. (2) Step 2, automated building fire risk assessment, combines the information extracted from the previous step to evaluate the interconnectivity of each spatial segment. The emphasis in this step is on determining room-to-exit evacuation distances to assess fire risks. (3) Step 3, AI-enabled building fire risk mitigation, focuses on redesigning blueprints identified as high risk in step 2. This is accomplished by developing a deep generative model to modify and adjust the layout.

3.1. Automated polymorphic building blueprints interpretation

In step 1 of the proposed approach, diverse architectural blueprints are standardised to generate a uniform, computer-readable format. This step sets the foundation for the subsequent analysis. The diverse nature and varying standards of the original blueprints necessitate a robust pre-processing approach, which is critical for accurate interpretation. As illustrated in Fig. 2, this blueprint standardisation approach encompasses resizing, denoising, size reduction, binarisation, and morphological operations, which collectively work to enhance the blueprint image quality, reducing noise and amplifying critical features necessary for precise interpretation.

The initial stage of image processing involves resizing, which alters the width and height of the image. Gaussian filtering is utilised to decompose image frequencies using specified parameters like sigma and window size, enhancing the performance of edge-detection algorithms that are prone to noise interference. Following this, grey processing is employed to convert RGB images to greyscale, thereby averting potential distortion during size reduction. The binarisation technique then

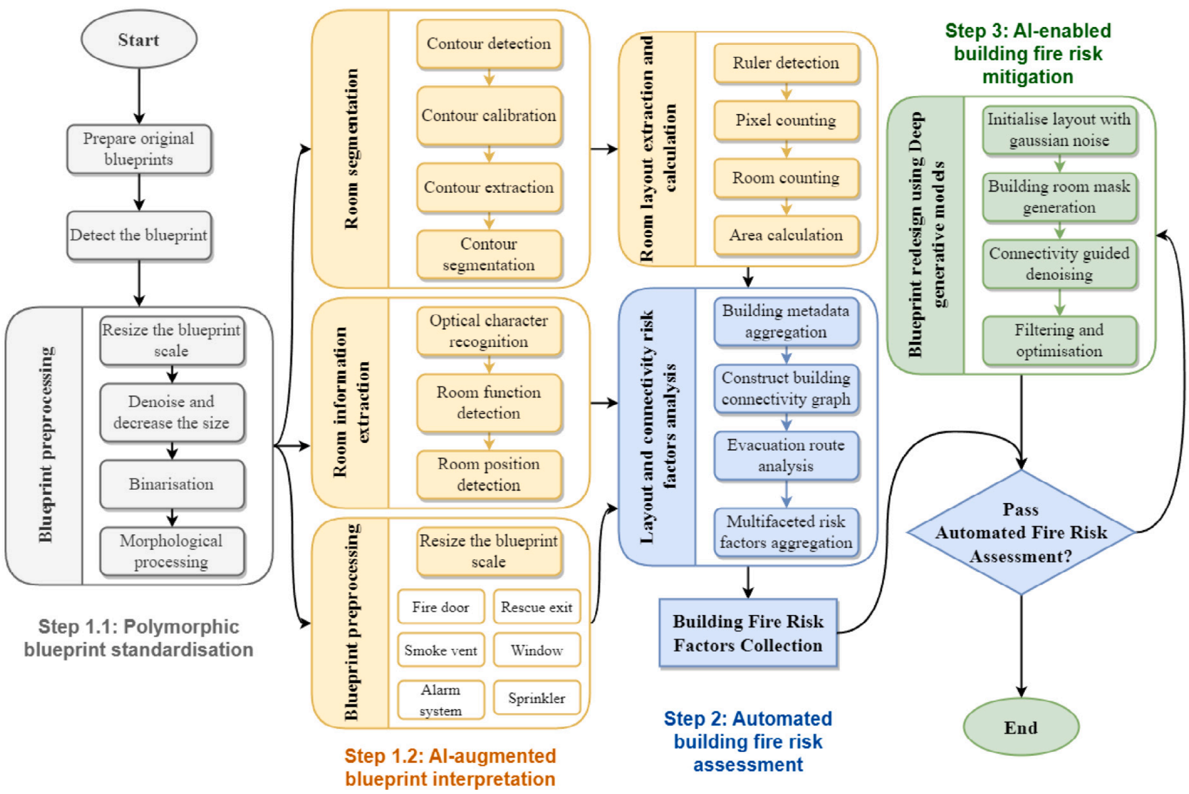


Fig. 1. The proposed approach of automated building fire risk assessment and mitigation.

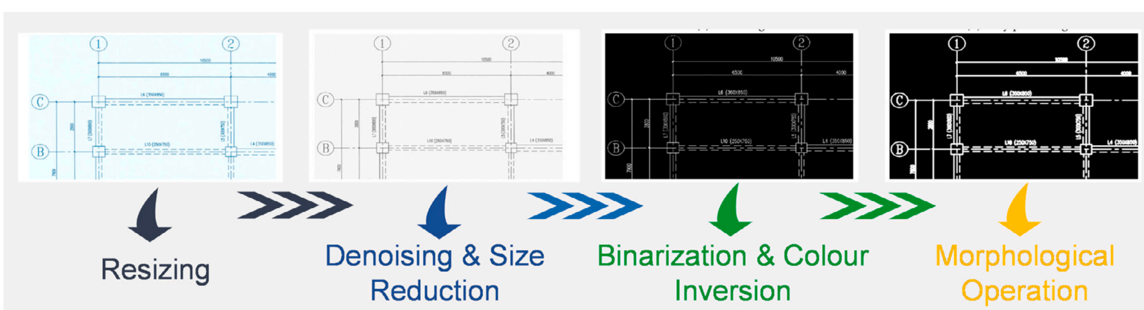


Fig. 2. Polymorphic blueprint standardisation approach.

converts the floorplan images to a black-and-white format, facilitating object features and edge extraction. Lastly, morphological operations are applied to refine image clarity and facilitate feature extraction.

With the standardisation completed, the proposed approach begins on a series of complex interpretation tasks, including room segmentation, layout extraction, room information extraction, and the identification of fire safety facilities including fire exits. These tasks are segregated into two primary categories: contour processing, which handles the macro-level interpretation of blueprint layouts, and room-related processing, focusing on the micro-level details within the blueprints.

Fig. 3 depicts the overall automated blueprint interpretation process, which commences with identifying fundamental contours that define the building's structure, employing a series of processing elements like contour detection, calibration, extraction, and segmentation. This is supplemented by colour detection, particularly for objects coded by colour, aiding in the comprehensive understanding of the blueprint. Techniques such as ruler detection and pixel counting are employed in calculating room dimensions and for determining evacuation routes. Subsequently, the system delves into capturing more nuanced information, including doors, fire exits, sprinkler systems, and textual data like room functions and dimensions. This is achieved through a combination of geometric recognition and advanced OCR algorithms.

During the interpretation process, various computer vision functions and algorithms have been employed. For instance, the Canny edge [70] detection function is crucial for outlining image contours through a series of steps, including noise reduction, gradient calculation, and edge tracking. The integration of vertical and horizontal line detection enables effective segmentation of the blueprint into distinct rooms. Furthermore, the Contrast Limited Adaptive Histogram Equalisation (CLAHE) [71] technique is employed to enhance the clarity of scanned images, especially useful for accurately interpreting scaled rulers and calculating precise room areas and evacuation distances. The use of both single-object and multi-object matching methods facilitates the extraction of specific entities, such as fire safety facilities, from the blueprints.

The outputs of the blueprint interpretation process comprise two fundamental elements. The first is a vectorised floorplan, providing a streamlined and consistent representation of the original layout. The second is an extensive set of enhanced blueprint metadata, encompassing crucial information like material specifications, room type

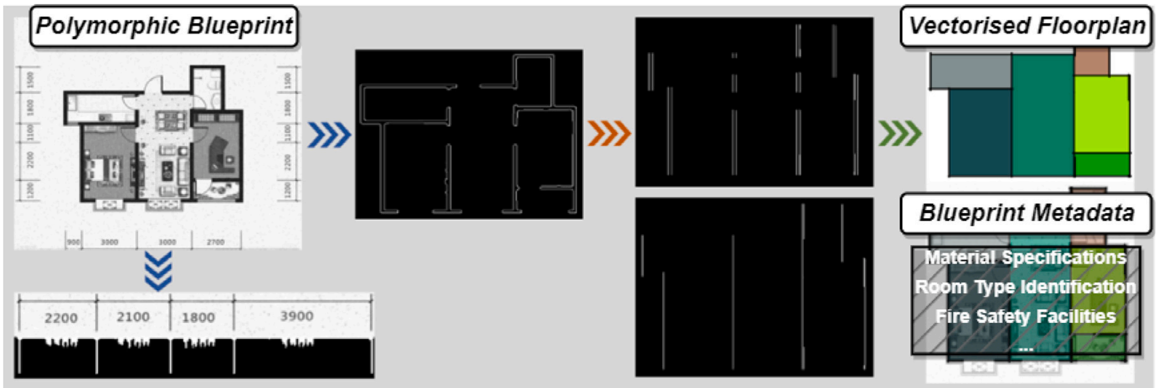


Fig. 3. Automated polymorphic building blueprints interpretation process.

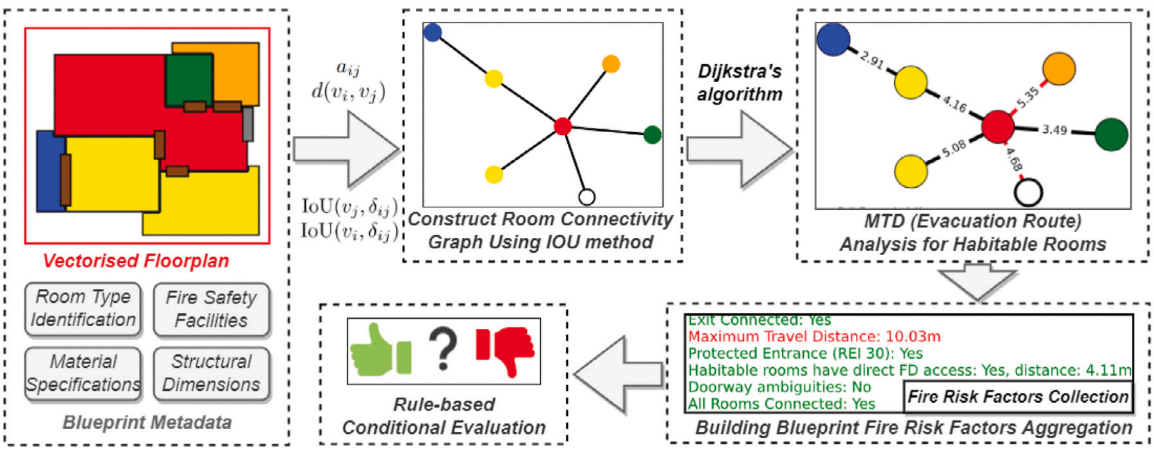


Fig. 4. Automated building fire risk assessment process.

identification, fire safety facilities, and structural dimensions. This integration of a standardised vectorised floorplan with detailed enhanced blueprint metadata lays a solid foundation for a practical and automated fire risk assessment, ensuring a comprehensive approach where relevant factors are considered in evaluating building fire safety.

Integrating a standardised vectorised floorplan with the rich enhanced blueprint metadata lays the groundwork for a practical and automated fire risk assessment. For further insights into the detailed methodological process of the standardisation and overall interpretation process, readers are directed to our previous work [20,72]. This comprehensive approach ensures that all relevant factors are considered when evaluating fire-related risks of a building.

3.2. Automated building fire risk assessment

Building on the standardised and well-interpreted blueprints from Section 3.1, essential parameters such as vectorised bounding boxes for room layouts, floor height, available emergency evacuation means (e.g. windows and doors), and fire resistance levels of building materials, specifically their REI standards [73], a quantitative fire resistance rating, have been acquired. This subsection elucidates the knowledge-based algorithm driving the automated building fire risk assessment.

The proposed fire risk assessment criteria largely draws from the UK Building Regulations 2010, specifically the Approved Document B (fire safety) volume 1: Dwellings, 2019 [19]. This document elucidates how floor placement within a building impacts standards (sections 2.4–2.6), provides specifications for compliant emergency escape windows and doors (section 3.6), discusses the permissible interconnectivity of habitable rooms and the allowance of inner rooms (sections 3.7–3.8), defines precise MTD lengths (section 3.18 b), and outlines conditions under which the distance from a room door to a fire exit can replace traditional MTD measurements (sections 3.19 and 3.18 a).

It is crucial to understand that the scope of this approach is to validate the practicality of automated fire risk assessment directly for building blueprints. This entails showcasing the approach's capabilities in automating the fire risk assessment process for building layout blueprints and establishing its practical applicability within the context of UK Building Regulations, addressing geometric and topological aspects of the design, particularly the MTD criterion. This research is not intended to encompass a full-scale building code compliance check but to demonstrate a capable and flexible automated fire risk assessment method within a specific scope of building regulations.

Fig. 4 depicts the overall automated fire risk assessment process. To craft the requisite automated fire risk assessment algorithm, foundational blueprint information garnered in step 1 is utilised. This paves the way for advanced analysis, which enables the computation of MTD and the determination of the presence of inner rooms.

All the spaces and distances within a building are abstracted and integrated into a weighted undirected graph, resulting in a room connectivity graph, G. This graph can be mathematically represented as $G = (V, E)$, where $V = \{v_i\}_{i=1}^{N}$ signifies the ensemble of rooms and fire exits (an exit is considered an external room), and $E = \{e_{ij}\}$ represents the connectivity between rooms—if present, it indicates that two vertices (rooms or exits) are linked. In this representation, N indicates the aggregate number of rooms and exits. Typically, v_N is synonymous with the singular exit of a building. The connection, e_{ij}, denotes the pairwise linkage between room v_i and room v_j.

Each room, v, while serving as a vertex in the connectivity graph, also encapsulates the geometric attributes of the space it represents.

This dual functionality is achieved by associating each v with a bounding box, defined by a collection of D corner coordinates in a two-dimensional space, formalised as $v \in \mathbb{R}^{D\times 2}$. Here, D refers to the count of corners defining the room space. To exemplify, for a square room, D equals four.

To infer room connectivity, bounding boxes (encompassing rooms, doors, and exits) derived from vectorised blueprints are assessed using the Intersection over Union (IoU) method. This evaluation facilitates the automated identification of connections (i.e., doors) between each room and its adjacent rooms or exits. δ_{ij} refers to the door between rooms v_i and v_j, with doors themselves being exempt from the graph model G. A door, δ, connecting two rooms is inferred if the IoU of δ with both rooms is greater than zero and holds the highest IoU value compared to any other room.

The comprehensive interconnections of all rooms and exits are encapsulated in the adjacency matrix $A \in \mathbb{R}^{N\times N}$. Each element a_{ij} in this matrix is a floating-point number, quantifying the distance between room spaces v_i and v_j. This distance is approximated by computing the direct linear distance, d, between the centroids of bounding boxes of the room spaces. This resulting value d is integrated into the adjacency matrix A using the mathematical expression:

$$a_{ij} = \begin{cases} d(v_i, v_j) & \text{if } \mathrm{IoU}(v_i, \delta_{ij}) > 0 \wedge \mathrm{IoU}(v_j, \delta_{ij}) > 0 \\ & \forall k \left(\mathrm{IoU}(v_k, \delta_{ij}) \le \mathrm{IoU}(v_i, \delta_{ij}) \wedge \mathrm{IoU}(v_k, \delta_{ij}) \le \mathrm{IoU}(v_j, \delta_{ij})\right) \\ & i \neq j, \\ 0 & \text{otherwise}, \end{cases} \tag{1}$$

Employing the connectivity graph structure, an analysis is applied to this undirected graph as detailed in Algorithm 1, to check whether all habitable rooms are at most one room away from the exit. The algorithm initialises by defining a set HR containing all habitable rooms, and for each room v in HR, employs a breadth-first search (BFS) strategy to explore paths from v to v_N. During the BFS, a queue Q and a visited set $visited$ are initialised, adding the current room v with its depth (initially 0) as a tuple to the queue. BFS expands the search layer by layer, ensuring nodes are processed in proximity order, and continues until an exit is found or it is confirmed that the exit cannot be reached within a two-room limit, ultimately outputting a boolean to signify the presence of inner habitable rooms that meet the specified connectivity criteria.

Furthermore, each room's theoretical escape route is framed as a shortest-path problem, for which Dijkstra's algorithm [74] is employed. This algorithm uses a greedy strategy to iteratively select the nearest unvisited node and update the distances to adjacent nodes, thereby calculating the escape distances for all habitable rooms. Moreover, for designs adhering to specific criteria—including REI 30 standard for a protected hall, compliance of room doors with fire door standards, and a direct access provision from the room door to the hall—the MTD can be exempted and replaced by the furthest linear distance from any habitable room door to the fire exit. Fire risk factors, such as MTD and inner rooms, are retrieved by employing the aforementioned interconnectivity graph analysis methodologies on these building blueprints. Combining these factors makes up a fire risk factors collection used for automated regulation compliance checking.

Towards the end of the automated fire risk assessment process, a rule-based conditional evaluation, as illustrated in Fig. 5, is employed to determine whether the building blueprint meets the requisite fire risk regulatory criteria. This conditional evaluation process is based on a series of 'yes' or 'no' decisions on the fire risk factors, including the value of MTD and the presence of inner habitable rooms. The final outcome of this evaluation is represented as either a 'Pass' or 'Fail' verdict, thereby providing a clear and decisive assessment of the building's compliance with fire safety norms.

Algorithm 1 Inner Habitable Room Checking Algorithm

Require: An undirected graph G with vertices V and edges E.
Ensure: Boolean indicating whether all habitable rooms in V are at most one room away from the exit v_N.

```
Let HR ⊆ V be the set of all habitable rooms in G.
for each room v in HR do
    if v ≠ v_N then
        if not validConnectionToExit(v, G) then
            return False
        end if
    end if
end for
return True
function validConnectionToExit(v, G)
    Initialise a queue Q to empty
    Let visited be an empty set
    Enqueue (v, 0) into Q
    return BFS(Q, visited, G)
end function
function BFS(Q, visited, G)
    while Q is not empty do
        (current, depth) ← dequeue from Q
        if current = v_N then
            return True
        end if
        if depth < 2 then
            for each neighbor of current in G do
                if neighbor not in visited then
                    Add neighbor to visited
                    Enqueue (neighbor, depth + 1) into Q
                end if
            end for
        end if
    end while
    return False
end function
```

3.3. AI-enabled building fire risk mitigation

The proposed fire risk mitigation approach leverages advanced deep generative models, particularly diffusion models, to redesign unqualified architectural floorplans. Employing the room connectivity graph from the original design as a constraint, the model optimises the MTD to ensure safety compliance while adhering to room types and counts of the original design specified in the constraint.

To conceptualise the generative task of redesigning blueprints within the proposed fire risk mitigation approach, a set of bounding boxes encompassing all spaces within the house is introduced, denoted as B. To elaborate, $B = \Delta \cup V$, where Δ represents a set of doors δ and V stands for a set of rooms and the exits. The room connectivity graph G of the original building layout serves as a condition to guide the mitigation deep generative model so that the generated floorplans align with the room types and counts of the initial design.

Diffusion models operate through a sequence of gradual transformations between data and noise distribution. This process is mathematically formulated in two main stages: the forward diffusion process (adding noise) and the reverse diffusion process (denoising) [66]. Guided Diffusion introduced classifier guidance to condition the diffusion models to enhance the control over training and generation phases [67].

The forward diffusion process gradually adds noise to the original data B_t to transform it into a Gaussian noise distribution over a sequence of T steps. A Markov chain can represent this process where each step, t, adds a small amount of Gaussian noise to the data. The process is designed so that B_T is nearly indistinguishable from pure noise. The reverse diffusion process aims to learn the reverse of the

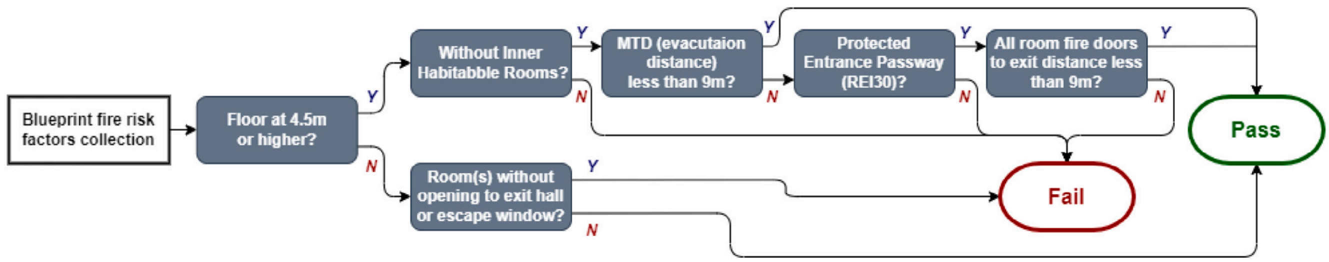


Fig. 5. Rule-based conditional evaluation on the fire risk factors.

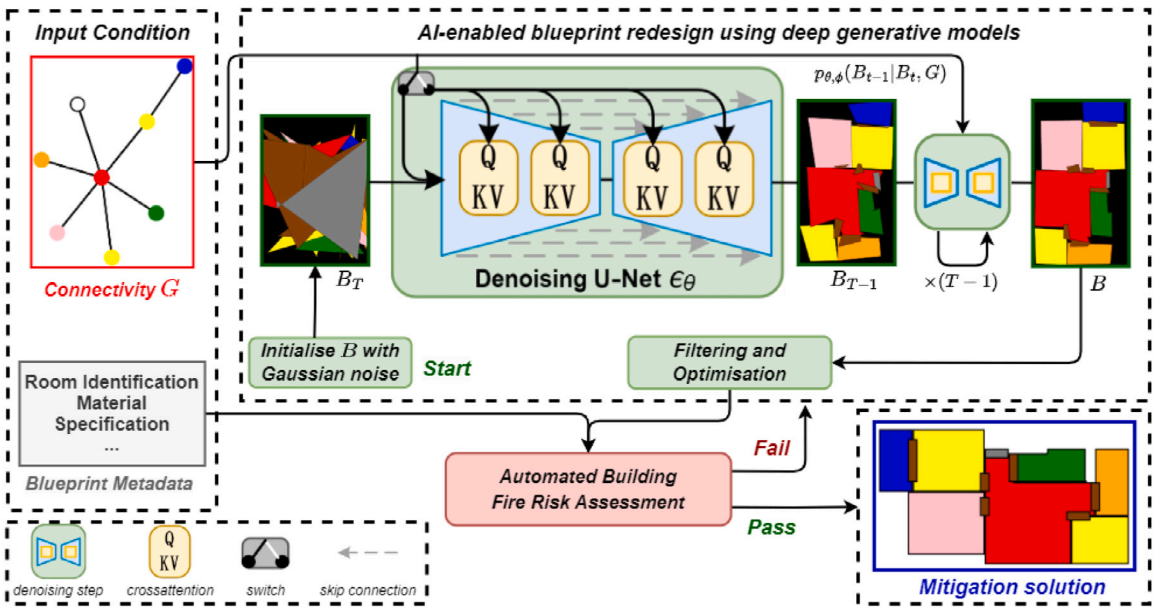


Fig. 6. AI-enabled building fire risk mitigation.

forward process, starting from noise and progressively removing it to reconstruct the data. It involves training a neural network model p_θ to estimate the original data B_0 from the noise B_T. This is achieved through stepwise denoising, where the model functions close to a series of denoising autoencoders:

$$\epsilon_\theta(B_t, t); t = 1, \ldots, T \tag{2}$$

here, the autoencoders $\epsilon_\theta(B_t, t)$ are implemented through Transformer-based [75] U-Net architectures [76] and are trained to predict the denoised form of their input B_t.

The guided diffusion model is utilised in the adopted approach, and the room connectivity graph G is employed as the guided condition. B is initialised with Gaussian noise and the room connectivity graph G. Using G as the condition, the model continuously de-noises to automatically generate B with the complete bounding box vector coordinates; this continuous reverse process can be summarised as:

$$p_{\theta,\phi}(B_{t-1}|B_t, G) = Z p_\theta(B_{t-1}|B_t) p_\phi(G|B_t) \tag{3}$$

here, the Z is a normalisation constant, $p_\phi(G|B_t)$ denotes the condition classifier.

The proposed guided diffusion model for risk mitigation was developed based on the work of Shabani et al. [68]. The adopted model was trained on 50,000 floorplans sourced from the RPLAN dataset [69]. After processing and standardisation, as detailed in Section 3.1, parts of the techniques described in Section 3.2 were applied, especially the formula (1), to construct the room connectivity graph G. The model was trained with G as the condition guiding the diffusion model to generate building floorplan blueprints that adhere to the respective room connectivity layout.

Fig. 6 depicts the comprehensive workflow of the proposed AI-enabled fire risk mitigation process. Upon employing the diffusion model, initially randomised room vectors are refined through a continuous denoising process, guided by G. This approach ensures that the redesigned blueprints meet fire safety criteria and align with the room types and counts of the original design. Generated blueprints are subsequently filtered and assessed for fire risk using methods from Section 3.2. Additional filterings were implemented to improve blueprint aesthetics and design rationale. Those not meeting the criteria are discarded, prompting a reiteration of the redesign process. This iterative mechanism ensures that only those blueprints that successfully pass the fire risk assessment while maintaining a level of adherence to the original design intent are finalised as mitigation solutions.

4. Implementation and case study

This case study aims to demonstrate the applicability and effectiveness of the proposed approach in interpreting, evaluating, and mitigating fire risks from residential building blueprints. This case study focuses on residential buildings due to the prevalence of accessible public datasets and the significant impact that improved fire risk assessment can have on enhancing safety within such environments. This case study utilised public data sources of residential floorplans and a selected set of blueprints of real residential buildings. The public dataset, notably the RPLAN dataset, provided a comprehensive base of floorplan designs, while the inclusion of blueprints of real residential buildings aimed to enrich this base with additional metadata

Table 4
Python libraries and packages used in the proposed approach.

Library	Feature
Imageio [77]	A library for reading and writing images in various formats
Matplotlib [78]	A plotting library for creating static, interactive, and animated visualisations
Networkx [79]	A package for creating complex networks of nodes and edges
Numpy [80]	A fundamental package for scientific computing with Python
OpenCV [81]	An open source computer vision and machine learning software library
Pillow [82]	Adds image processing capabilities to the Python Imaging Library (PIL) fork
Pygraphviz [83]	A Python interface to the Graphviz graph visualisation package
Scikit-image [84]	A collection of algorithms for image processing in Python
Scipy [85]	A library for scientific computing and technical computing
Tesseract-OCR [86]	An OCR engine for various operating systems, wrapped into Python via Pytesseract
Pytesseract [87]	An OCR tool for Python, based on the Tesseract-OCR engine
PyTorch [88]	An open source machine learning library

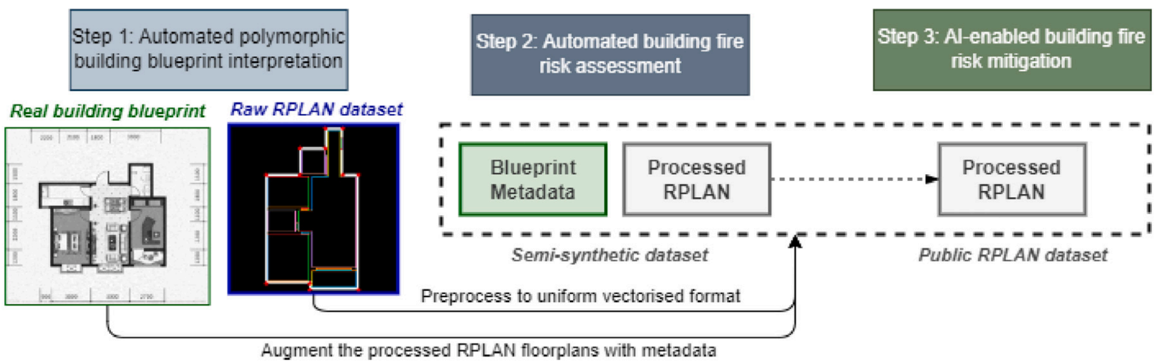


Fig. 7. Data usage for the implemented case study.

necessary for a thorough fire risk assessment. The development of a semi-synthetic dataset enabled detailed analysis of fire risk assessment across varied residential layouts.

The proposed approach was implemented in Python, leveraging a suite of libraries and packages outlined in Table 4. The case study was developed using Anaconda platform with Intel Xeon CPU and NVIDIA Tesla T4 GPU.

4.1. Automated polymorphic building blueprint standardisation and interpretation

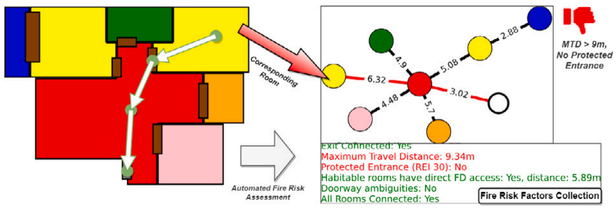
The first step focuses on the standardisation and interpretation of building blueprints, which was conducted in alignment with methodologies and computer vision models developed in our previous works [20,72], as also outlined in Section 3.1. These methods were implemented to process a select number of real residential blueprints. This was not a large-scale analysis but a focused extraction from a manageable subset of blueprints, chosen for their relevance and representation of common building features. The core objective here was to extract the blueprint metadata, which included material specifications, room type identification, fire safety facilities, and structural dimensions. These elements are essential for assessing fire risks and compliance checks against the proposed assessment criteria drawn from the UK Building Regulations. For instance, material specifications help identify the use of fire-resistant materials in crucial areas like protected entrances and fire doors. Room type identification is critical for determining which rooms are habitable, as regulations specify that habitable rooms should not be inner rooms. The analysis of fire safety facilities focuses on clearly defined fire exits. At the same time, structural dimensions, including floor height, are essential for calculating the distance from rooms to exits and can influence the assessment outcomes. Simultaneously, 50,000 floorplans from the publicly available residential floorplan dataset, RPLAN [69], were processed, transforming them into a uniform vectorised format. This transformation was essential for training and evaluating the deep generative model later. However, these floorplans from RPLAN lacked the relevant blueprint metadata.

As depicted in Fig. 7, a semi-synthetic dataset was formed by processing publicly available floorplans from the RPLAN dataset and augmented with metadata extracted from a selection of real residential blueprints. The augmentation process involved randomly assigning extracted metadata to the vectorised processed RPLAN floorplans, thereby enriching them with essential building features. This synthetic dataset offers general design information in the form of vectorised floorplans and the fire risk-related building metadata necessary for a comprehensive fire risk assessment.

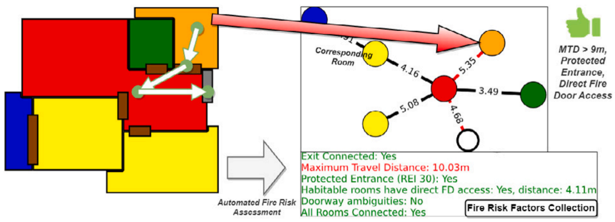
4.2. Automated fire risk assessment

In verifying the automated fire risk assessment and the developed deep generative model, the proposed fire risk assessment process utilises this synthetic dataset, combining the enhanced metadata with the RPLAN floorplans to conduct a thorough evaluation. The assessment adheres to parts of the rules outlined in the UK Building Regulations 2010, Approved Document B (fire safety) volume 1: Dwellings, 2019 edition. One of the critical criteria evaluated is the MTD, as well as a focus on the compliance of room doors with fire door standards and the presence of a direct access provision from the room door to the entrance. In cases where specific conditions are met, such as adherence to the REI 30 standard for a protected entrance and fire doors present, the MTD requirement can be exempted and replaced by the linear distance from any habitable room door to the fire exit.

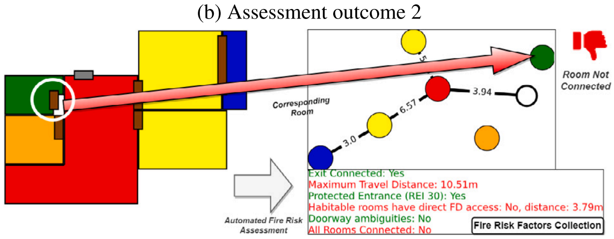
Fig. 8 showcases three examples of the results from the implemented Automated fire risk assessment with additional design rationality filtering. The assessments leveraged randomly assigned blueprint metadata, particularly focusing on REI ratings to determine the compliance of floorplans with fire safety standards. For example, Figs. 8(a) and 8(b) did not meet MTD requirements, exceeding the maximum distance of 9 m. However, Fig. 8(b), with its REI 30 Protected Entrance and compliance with fire door standards, passed the assessment as the MTD



(a) Assessment outcome 1



(b) Assessment outcome 2



(c) Assessment outcome 3

Fig. 8. Automated fire risk assessment outcomes.

criteria were replaced by the furthest linear distance from any habitable room door to the fire exit. Additional filtering was implemented to enhance blueprint aesthetics and design rationale. Blueprints with ambiguous or missing room doors, resulting in incorrect connectivity to the overall graph, were discarded, as shown in Fig. 8(c).

4.3. AI-enabled building fire risk mitigation

This implementation utilised PyTorch to implement the deep generative model based on the work of Shabani et al. [68] for blueprint risk mitigation. The 50,000 processed RPLAN floorplans were employed for the training process. Following the methods in Section 3.2, room connectivity graphs G for each floorplan were constructed. The model was trained using the Adam [89] optimiser with a learning rate of 0.001 on an NVIDIA Tesla T4 GPU. Each floorplan's G served as the condition, guiding the diffusion model to generate designs adhering to the respective G.

Fig. 9 displays the post-training redesign outcomes of the implemented risk mitigation diffusion model. These designs, which successfully passed the automated fire risk assessment and additional filtering, exhibit diversity while meeting fire safety requirements. These designs also aligned with the room types and counts of the respective original designs, demonstrating that risk mitigation and design consistency are essential in the proposed approach.

5. Discussion

This research introduces an innovative, automated fire risk assessment and mitigation approach explicitly tailored to building blueprints. Distinct in its approach, the study focuses on processing raw 2D floorplan blueprints, a significant advancement over traditional methods that predominantly rely on structured BIM data. The relevance of this study is underscored by its ability to bridge the gap for buildings lacking detailed BIM data, thereby broadening the scope of automated code compliance and fire risk assessment to include less structured data sources.

Compared to previous methods, the developed approach is overperforming in its processing efficiency by providing a more streamlined solution, system capability by delivering a full-cycle approach to assess and mitigate building fire risks, data accessibility by utilising public datasets for developing the proposed approach, and system flexibility by its applicability for a wide range of buildings with or without BIM data.

5.1. System robustness and efficiency

5.1.1. Robustness to handle image noises

The developed approach is initiated by interpreting raw building blueprints. However, the quality of these blueprints varies due to factors such as lighting conditions, ageing of the documents, creases, and degradation during digital transmission. The effects of image noises on the blueprint interpretation process are crucial as they impact the effectiveness of subsequent fire risk assessment and mitigation algorithms. Consequently, assessing the robustness of the blueprint interpretation algorithm against different noise distributions is of interest.

To simulate real-world noise conditions, clear blueprint images were artificially altered by introducing two common types of image noise: Gaussian and salt-and-pepper noise. Gaussian noise, typically a result of electronic and optical noise in the imaging sensors, especially prevalent under low light conditions [90], is introduced by adjusting the sigma levels to vary the noise intensity. Conversely, salt-and-pepper noise, which simulates scenarios where digital images are degraded by sharp disturbances, such as transmission errors or faulty image sensors [91], is introduced similarly to test its impact. The effectiveness of the noise-introduced images was evaluated using the Peak Signal-to-Noise Ratio (PSNR), a quantitative measure of image quality. Generally, PSNR values above 40 decibels (dB) indicate excellent image quality, while values below 20 dB suggest significant degradation [92]. The tests covered a range of noise levels from 40 dB to 13 dB.

Fig. 10 displays the effects of added noise on floorplan layout detection, specifically focused on room segmentation. The algorithm processes the images to recognise the layout and identify room areas, consistently identifying six rooms correctly until the PSNR drops to 19 dB. Below 18 dB, however, the accuracy declined, with misidentifications and failures to recognise certain rooms occurring due to noise interference with the algorithm's ability to detect wall pixels correctly. Images with Gaussian noise showed a more rapid decline in room recognition, dropping to three correctly identified rooms and missing some entirely. Fig. 11 presents the detection accuracy of blueprint metadata, specifically the structural dimensions. Up to an PSNR of 22 dB, both types of noise-affected images retained over 95% accuracy. Unlike layout detection, the decline in accuracy due to salt-and-pepper noise was more pronounced, indicating a greater interference with OCR capabilities. The detection accuracy dropped significantly when the noise level exceeded 16 dB, indicating a substantial impact on OCR performance.

To summarise, the developed approach demonstrates robustness against Gaussian and salt-and-pepper noise at levels above 20 dB, effectively processing blueprints whose quality is somewhat compromised. However, more severe noise levels significantly reduce the performance of the blueprint interpretation algorithm. Improving the resistance of the layout detection algorithm to Gaussian noise and the OCR algorithm to salt-and-pepper noise is recommended for future works.

5.1.2. Efficiency of processing

Furthermore, comparing efficiency between the developed approach and established BIM-based methods, such as those represented by Fitkau et al. [10] and Malsane et al. [8], reveals marked distinctions.

Table 5 provides an estimated comparison of the time required for each phase of the compliance checking process using both the

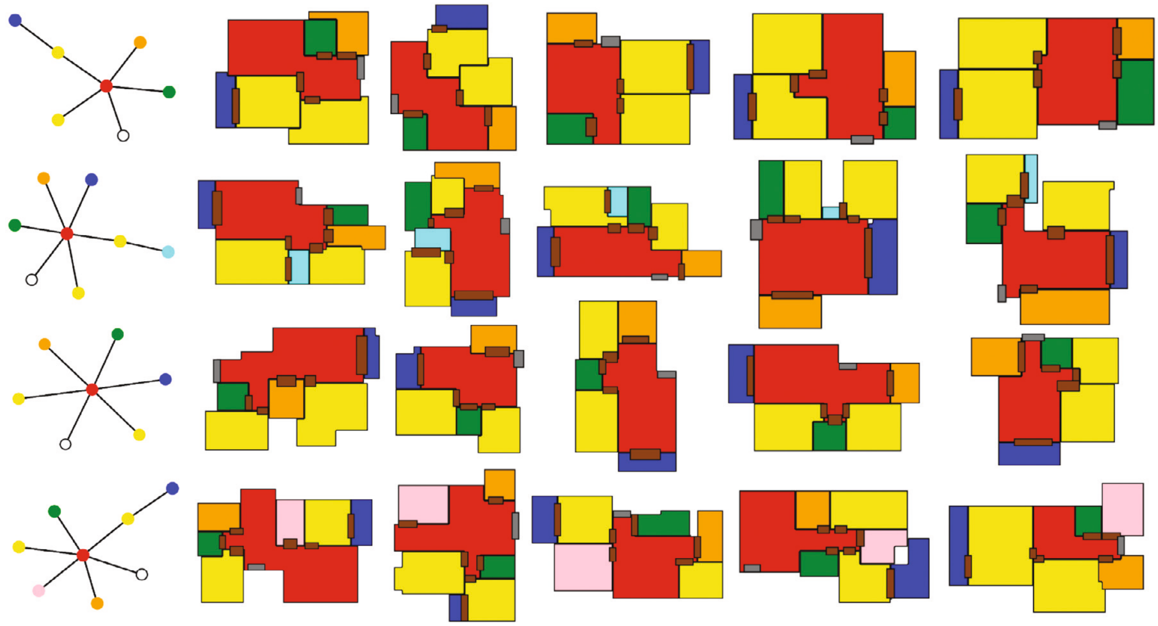


Fig. 9. Risk mitigation diffusion model redesign outcomes.

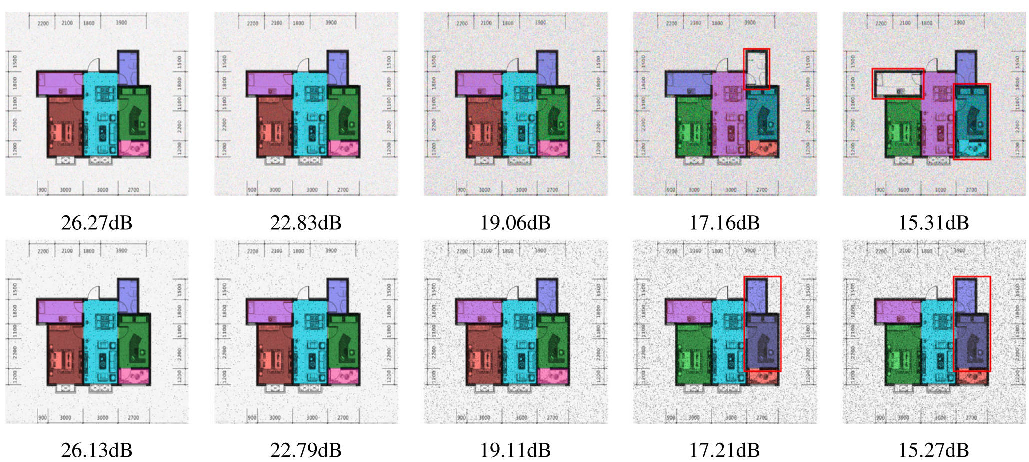


Fig. 10. Layout detection results with varied noise levels; mis-segmented rooms highlighted by red bounding boxes.

Table 5
Comparative efficiency analysis of automated compliance checking approaches.

Phase	BIM-based Method [10]	Developed Approach
Data Collection and Preparation	3h–10h (building BIM model)	~ 60 s (blueprint processing)
Risk and Compliance Checking	5–10 min (IFC file importing and reasoning in Protégé)	~ 2 s (direct execution)
Risk Mitigation	N/A (manual methods only)	~ 40 s (using deep generative models)

BIM-based approach and the developed approach. Previous BIM-based methods require the input of a BIM model for automated compliance checking, which involves substantial resource and time investment, particularly for buildings that do not already possess such models. BIM modelling for large-scale projects can take months of work for a professional engineer [93]. The residential properties, as the focus of this work, although consisting of smaller areas, can still take a considerable time to model. For instance, semi-manually creating a BIM model with the aid of a Lidar system for three as-built single-room structures of approximately 50 square metres can take up to 200 min, as noted by Li et al. [94]. Once the BIM model is constructed, it must be converted into an IFC file format and imported into Protégé for further analysis, adding additional steps and time to the process.

In contrast, the developed approach initiates automated compliance checking directly from architectural blueprints without the need for manual BIM model creation. This approach markedly reduces the preparatory phase, requiring approximately 60 s for blueprint preprocessing. Furthermore, the risk and compliance checking phase of the developed approach, directly executed without needing file conversion, takes only about 2 s. In addition to the streamlined data collection and risk assessment processes, the developed approach also introduces an automated risk mitigation process, which is absent in previous methods. Utilising deep generative models, the developed system can perform risk mitigation tasks in approximately 40 s.

Overall, the developed approach significantly outperforms the existing BIM-based method in terms of efficiency, providing a more

Table 6
Comparative analysis of system capability.

Criteria	Developed Approach	Fitkau et al. [10]	Zhang et al. [53]	Bloch et al. [54]	Li et al. [12]
Preprocessing	Yes, utilises raw 2D blueprints	Absent, structured data required	Yes, building-code corpora preprocessing	Absent, structured data required	N/A (post-hazard focus)
Risk Assessment	Yes, connectivity graph-based approach	Yes, SWRL rule-based analysis	Yes, risk inference using semantic representations	Yes, using graph neural network classifier	No direct risk assessment
Mitigation Strategies	Yes, AI-driven layout redesign	Not addressed	Not included	Not addressed	Yes, evacuation planning
Fire Risk Coverage	Yes, with focus on blueprint analysis	Yes, with focus on structural fire safety	General compliance, potential for fire risk	General compliance, potential for fire risk	Yes, specific to multifloor building evacuation
Full-cycle Capability	Yes, from preprocessing to mitigation	Risk assessment only	Risk assessment only	Risk assessment only	Mitigation-focused

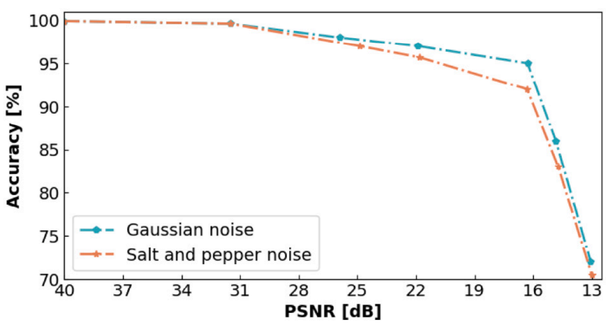


Fig. 11. Blueprint metadata OCR accuracy with varied noise levels.

streamlined and less resource-intensive solution. This is particularly advantageous in contexts where architectural data is not readily available in structured BIM models.

5.2. System capability

The comparative analysis of previous approaches, as depicted in Table 6, underscores the unique capabilities of the developed approach in the automated building risk assessment and mitigation landscape. A critical examination reveals that, unlike many existing studies, this approach offers a comprehensive full-cycle solution encompassing preprocessing, risk assessment, and mitigation steps. This end-to-end approach distinguishes it significantly from other research in the field, which primarily focuses on risk assessment without extending to mitigation solutions.

A key aspect where the developed approach stands apart is its preprocessing capability. Unlike studies that require structured data inputs, this approach is adept at processing raw 2D floorplan blueprints. This preprocessing stage is fundamental, as it allows for the application of advanced AI techniques to a wider range of properties, many of which may not have detailed BIM models. The ability to work with such raw data sources considerably enhances the approach's applicability and utility in real-world scenarios.

For risk assessment, the developed approach not only aligns with but also extends the capabilities of its counterparts. While studies like that of Fitkau et al. [10] utilise SWRL rule-based inference for risk and compliance assessment, they do not encompass the subsequent mitigation step. The proposed approach's integration of risk assessment with AI-driven mitigation techniques is a novel contribution, enabling it not only to identify potential fire risks but also to offer practical solutions for redesigning building layouts to mitigate these risks.

Moreover, the developed approach's coverage of fire risk is comprehensive, addressing both the immediate assessment needs and providing long-term mitigation strategies. This dual focus on assessment and mitigation is absent in other studies, which either concentrate solely on risk assessment or, as in the case of Li et al. [12], focus on post-hazard mitigation without a preceding risk assessment phase.

In essence, the system capability comparison highlights the developed approach's comprehensive nature, innovatively benchmarking the automated fire risk assessment and mitigation. Its full-cycle approach, from preprocessing raw blueprints to delivering AI-enabled mitigation solutions, marks a significant leap forward, filling gaps left by existing research and paving the way for more integrated and holistic automated building safety and compliance solutions.

5.3. Data accessibility for system development

The comparison of data accessibility and usage, as detailed in Table 7, plays a pivotal role in understanding the practical implications of different approaches to fire risk assessment and mitigation. The developed approach's reliance on publicly accessible datasets, specifically the RPLAN dataset [69] stands in contrast to other studies that either depend on private datasets or require data that is not readily replicable.

For instance, the works of Fitkau et al. and Zhang et al. utilise sources of data that are not easily accessible, or not readily available for replication or extending the work to broad applications [10,53]. Fitkau et al. formed their study on building fire safety ontology knowledge model, based on expert knowledge extracted from various sources, including semi-structured interviews, which poses challenges in terms of replicability due to the confidential and expert-dependent nature of the information. Zhang et al. processed and labelled semantically annotated building-code requirements from specific building codes, which, while public, involve complex preprocessing work to build the data foundation.

In contrast, the proposed approach's use of open, publicly available datasets with large volume of data exemplifies high accessibility, contributing to the ease of replication and scalability of the study. The use of vectorised building blueprint floorplans from a publicly available dataset enhances not only the approach's applicability but also positions it as a more inclusive tool, capable of catering to a broader range of architectural scenarios, especially those lacking in detailed BIM models.

The data accessibility aspect of the proposed approach thus underscores its potential for widespread application and adaptability. By harnessing publicly available data, the approach demonstrates a pragmatic approach to building safety solutions, making it an invaluable tool in the realm of fire risk assessment, especially for properties that fall beyond the scope of traditionally BIM-based studies.

5.4. System flexibility

The flexibility of the developed approach, as discussed in Table 8, reveals its flexibility to different data formats and regulations, and broad application scope compared to other studies. This flexibility is primarily attributed to its capability to process raw 2D floorplan blueprint designs, which is a notable departure from the reliance on detailed BIM models seen in other research.

In the context of data format requirements, conventional automated compliance checking methods, like those conducted by Fitkau et al. and Zhang et al. necessitate IFC-based BIM models [10,53]. While

Table 7
Comparison of data accessibility for system development.

Criteria	Developed Approach	Fitkau et al. [10]	Zhang et al. [53]	Bloch et al. [54]	Li et al. [12]
Source of Data	Public dataset	Interviews with domain experts	Building codes (IBC, IECC, ADA Standards)	Synthetic, based on real-world floorplans	ABM simulations
Type of Data Used	Vectorised building blueprint floorplans	Expert knowledge to develop an ontology model	Semantically annotated building-code requirements	'Pass'/'Fail' labelled graph representations	Influencing factors and corresponding evacuation time
Data Volume	50,000 floorplan designs	Building codes and 7 semi-structured interviews	7500 sentences for training, 600 for testing	1000 synthetic designs	300 samples for training, 60 for testing
Availability	Publicly available	Confidential	Publicly available	Private to the specific study	Private to the specific study
Accessibility	High, use public data and clear processing method	Low, requires extensive expert knowledge	Average, publicly available regulations but with further selection process	Average, requires data synthesis	Average, requires setup of ABM simulations using proprietary software

Table 8
Comparison of system flexibility.

Criteria	Developed Approach	Fitkau et al. [10]	Zhang et al. [53]	Bloch et al. [54]
Data Format Required	Raw 2D floorplan blueprint designs with/without BIM models	IFC-based BIM model	IFC-based BIM model	Building geometric and topological data in specific format
Flexibility to Different Regulations	High, with adjustable parameters and rules	High, with adjustable parameters and SWRL rules	Average, with fixed rules extracted from given building code	Average, need to adjust the pass/fail labels in training dataset to alter classifier results
Application Scope	Broad, applicable to various types, including buildings with/without BIM models	Limited, only buildings with detailed BIM models	Limited, only buildings with detailed BIM models)	Potentially broad, can use wide range of building data but requires manual data extraction

such models offer comprehensive data for analysis, their requirement inherently limits the applicability of these frameworks to buildings where such detailed models are available. Conversely, the proposed approach's utilisation of raw blueprints significantly broadens its application range, making it suitable for a variety of building types, particularly buildings which might not have BIM data readily available.

The flexibility to different buildings is further accentuated by the approach's flexibility in terms of parameter adjustments and rule application. Unlike other studies that might require manual adjustments or offer limited flexibility in regulatory compliance, the proposed approach is designed to adjust parameters and rules for different regulations, enhancing its versatility across various architectural designs and safety standards.

Moreover, the application scope of the proposed approach is notably broad. While some studies like that of Bloch et al. [54] show potential in applying their methodologies to a wide range of data represented in graph format, the manual extraction of data poses a limitation. The proposed approach, with its ability to process less structured data formats, overcomes such limitations, offering a more accessible and user-friendly approach to fire risk assessment and mitigation.

In summary, the flexibility comparison highlights the proposed approach's strength in terms of its flexibility and wide-ranging applicability. This flexibility makes the approach a viable option for a diverse array of building types and underscores its potential to make significant contributions in areas where current research is limited, especially in providing fire safety and compliance solutions for buildings lacking comprehensive BIM data.

5.5. Future directions

The potential for future expansion and development of the proposed approach opens many ways for advancing automated building safety solutions in the designing and retrofitting stages. While the current implementation demonstrates significant efficacy in fire risk assessment and mitigation for buildings with only raw blueprint data, the foundational principles and methodologies established in this study have broader applicability.

One potential direction for future development is expanding the approach to encompass a more comprehensive range of building code regulations. The current focus on fire risk assessment, particularly within the constraints of raw 2D floorplan blueprints, sets a strong foundation. However, extending this approach to include additional aspects of building safety regulations could significantly enhance the approach's utility. Such an expansion would broaden the scope of risks and compliance issues that the approach can address and cater to a more diverse array of building types and design scenarios.

Another promising area for future development lies in the integration of the developed approach with emerging technologies and data sources. For instance, incorporating real-time data from Internet of Things (IoT) devices in buildings or leveraging advancements in AI and machine learning could further refine the risk assessment and mitigation processes. This integration could lead to more dynamic and responsive fire safety solutions, adapting to changing conditions within a building.

One of the fundamental aspects of the proposed methodology is the abstraction of different areas and spaces into nodes, representing the architecture internally as a graph. The proposed connectivity graph approach simplifies the complexity of architectural layouts into a computer-readable representation, demonstrating efficacy in calculating the MTD and evaluating other risk compliance factors such as connectivity and identifying inner habitable rooms.

As illustrated in Fig. 12, the proposed graph-based analysis methodology is applied to a complex office building blueprint featuring dozens of offices per floor. This complex connectivity can be efficiently extracted to calculate MTD and inner rooms for comprehensive fire risk assessment using the same automated techniques in the current approach. This graph-based approach holds potential for broader applications, such as incorporating real-time data from IoT devices in buildings and leveraging spatial–temporal AI models to monitor and predict the feasibility of live evacuation routes. For instance, temporary blockages in passageways due to various reasons can be dynamically integrated into the model. This approach paves the way for constructing a graph-based digital twin of the building, enabling real-time acquisition of interior data through multi-modal data fusion methods, such as

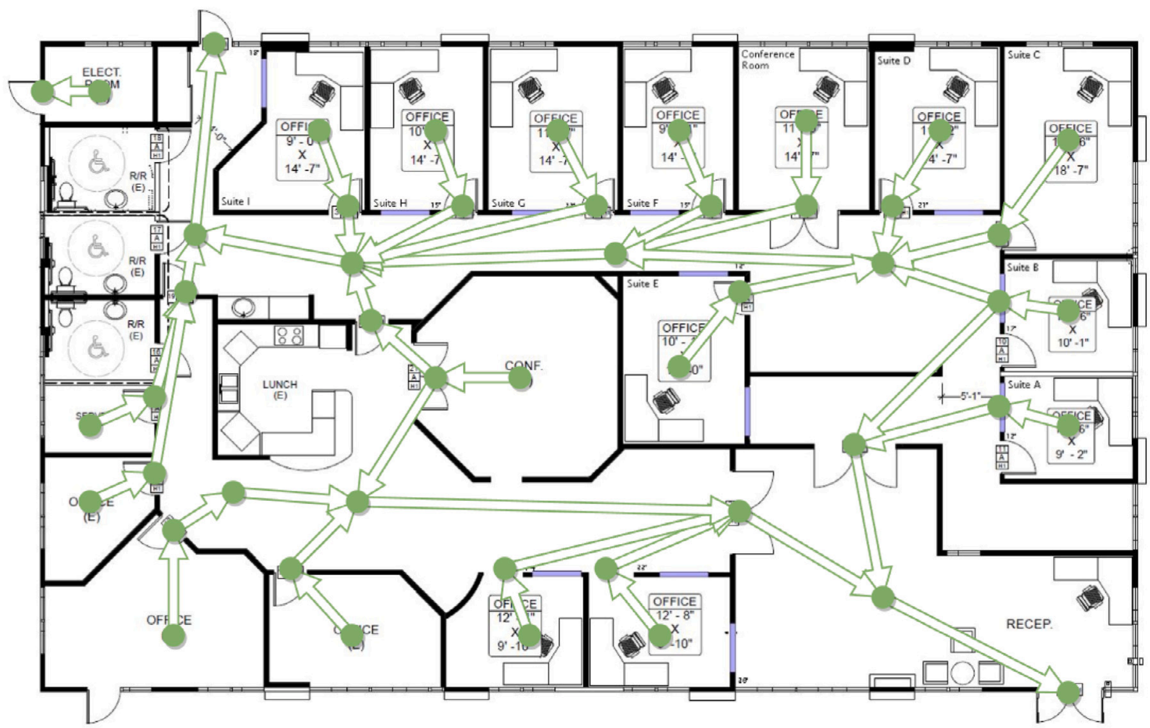


Fig. 12. Graph-based connectivity analysis in a complex office building.

IoT or camera-based analytics. Integrating such live data can transform the current static risk analysis into a dynamic, constantly updating risk evaluation model and a digital twin of building risk.

To summarise, with its innovative approach to fire risk assessment and mitigation, the proposed approach marks a significant advancement in the field and lays the groundwork for future developments. These expansions and integrations could further revolutionise the field of automated building risk and compliance assessment, making significant contributions to both individual building safety and broader urban risk management.

6. Conclusion

This paper has presented a comprehensive and innovative automated fire risk assessment and mitigation approach for building blueprints, employing advanced AI and deep generative models. It first adopted computer vision techniques to standardise and interpret building blueprints, extracting the vectorised floorplan layout and blueprint metadata containing critical building information. Then, an automated building fire risk assessment approach was developed to construct the room connectivity graph from the processed vectorised floorplan and evaluate fire risks in the building based on room interconnectivity and evacuation distances. Finally, it developed a deep generative model-enabled building fire mitigation approach to redesign building layouts and proactively address the identified risks. A case study has been investigated to verify the effectiveness of the developed approach on an application on residential buildings. Evaluations demonstrated that the blueprint interpretation algorithm exhibits satisfactory robustness to image noises, ensuring reliable analysis under varying conditions. The findings showed that the developed approach is adept at automating fire risk assessments directly utilising building blueprints by detecting potential risk factors and suggesting viable mitigation strategies, thereby offering a robust tool for enhancing building safety and compliance with fire safety regulations.

Compared to previous studies, a notable advancement is the approach's full-cycle capability, encompassing raw blueprint interpretation, risk assessment, and risk mitigation. Furthermore, integrated with AI-driven techniques, the developed approach can identify potential fire risks and proactively redesign building layouts to address these risks effectively. Its ability to rapidly process and accurately refine building layouts while considering both fire safety and design rationality, highlights its potential as a forward-thinking solution in architectural safety and fire risk compliance. The flexibility in processing less structured raw blueprint data enhances the approach's applicability across various building types. This adaptability is further exemplified by the approach's ability to abstract complex architectural layouts into a graph-based representation, simplifying intricate design information into an analytically manageable, and computer-readable graph representation. In addition, the enhanced processing efficiency, by eliminating the need for manual BIM model creation and IFC-file conversion, significantly reduces the time and resources required for automated compliance checking. The data accessibility of the developed approach, leveraging publicly available datasets like the RPLAN dataset, adds to its practical utility. Using such data can also ensure ease of replication and scalability, making this approach a more inclusive tool for fire risk assessment and mitigation.

This research contributes significantly to the field by providing a methodical approach to the automatic interpretation, risk assessment and mitigation using raw building blueprints. This study addresses a critical gap in automated fire risk assessment and compliance checking by providing methods to proactively address the identified risks and irregularities and effective automated fire risk management methods for buildings lacking structured and detailed BIM data.

The current implementation showcases notable effectiveness in automating fire risk assessments and mitigation strategies using building blueprints. However, its application is primarily confined to addressing topological and geometric aspects as dictated by sections of building code regulations. Future directions for this research include expanding the developed approach to cover a broader spectrum of building code regulations and incorporating a wider range of fire safety considerations, thereby broadening the scope of risks and compliance issues the approach can address. The established connectivity graph-modelling approach also lays a foundation to evolve the developed approach to a graph-based building risk digital twin by leveraging real-time data acquisition through multimodal data fusion techniques such as IoT sensors and camera-based analytics, transforming the static risk analysis into a dynamic, continuously updating model. These expansions and integrations could further revolutionise the field of automated building risk and compliance assessment, making significant contributions to both individual building safety and broader urban risk management.

CRediT authorship contribution statement

Dayou Chen: Writing – original draft, Validation, Software, Methodology, Investigation, Formal analysis, Data curation. **Long Chen:** Writing – review & editing, Supervision, Resources, Project administration, Methodology, Funding acquisition, Formal analysis, Conceptualization. **Yu Zhang:** Software, Methodology, Investigation, Formal analysis. **Shan Lin:** Writing – review & editing, Validation, Supervision, Resources, Data curation. **Mao Ye:** Writing – original draft, Software, Methodology, Formal analysis, Data curation. **Simon Sølvsten:** Writing – review & editing, Supervision, Resources, Project administration, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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